

Jacobian Descent For Multi-Objective Optimization

Valérien Rey — valerian.rey@gmail.com
Pierre Quinton — pierre.quinton@epfl.ch

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github.com/TorchJD/torchjd



torchjd.org



arxiv.org/pdf/2406.16232

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Theory

Partial order

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, as

$$\begin{aligned} \mathbf{u} \preceq \mathbf{v} \\ \Leftrightarrow \quad u_i \leq v_i \quad \forall i \in [m] \end{aligned}$$

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Examples:

$$\blacktriangleright \begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

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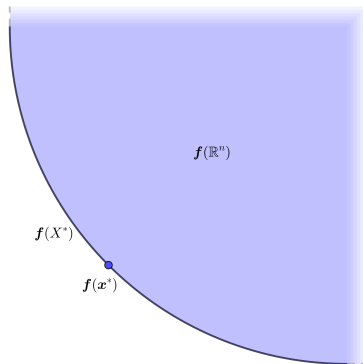
$$\begin{aligned} \mathbf{u} \preceq \mathbf{v} \\ \Leftrightarrow u_i \leq v_i \quad \forall i \in [m] \end{aligned}$$

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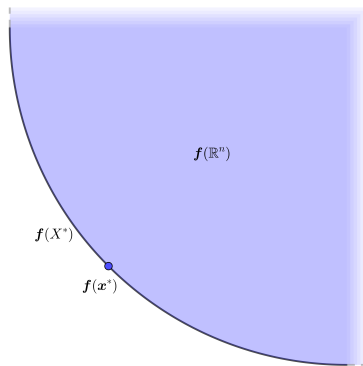
▶ $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

Pareto optimality



Pareto optimality

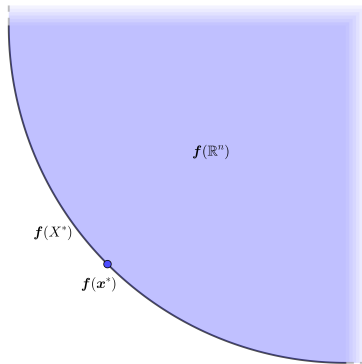
Pareto set: X^*



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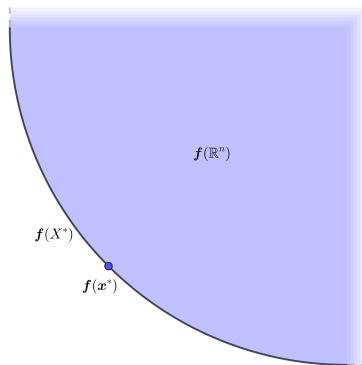
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Pareto front: $f(X^*)$



Pareto optimality

Pareto set: X^*
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$$\mathbf{x}^* \in X^* \quad \Leftrightarrow \quad \forall \mathbf{y} \in \mathbb{R}^n, \mathbf{f}(\mathbf{y}) \preceq \mathbf{f}(\mathbf{x}^*) \rightarrow \mathbf{f}(\mathbf{y}) = \mathbf{f}(\mathbf{x}^*)$$

Jacobian descent

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$$\mathcal{J}\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \nabla f_1(\mathbf{x})^\top \\ \nabla f_2(\mathbf{x})^\top \\ \vdots \\ \nabla f_m(\mathbf{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\mathbf{x}) & \frac{\partial}{\partial x_2} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\mathbf{x}) \\ \frac{\partial}{\partial x_1} f_2(\mathbf{x}) & \frac{\partial}{\partial x_2} f_2(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\mathbf{x}) & \frac{\partial}{\partial x_2} f_m(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\mathbf{x}) \end{bmatrix}$$

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We select $\mathbf{y} = -\eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x})) \in \mathbb{R}^n$.

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The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

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Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

Input: $\mathbf{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$

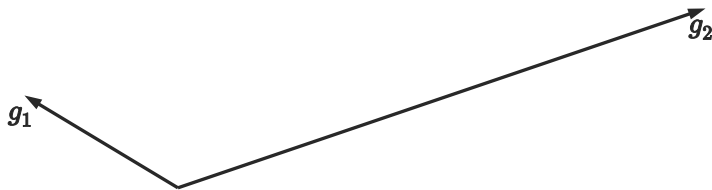
for $t \leftarrow 1$ **to** T **do**

$\mathbf{x} \leftarrow \mathbf{x} - \eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x}))$

Output: $\mathbf{x}$

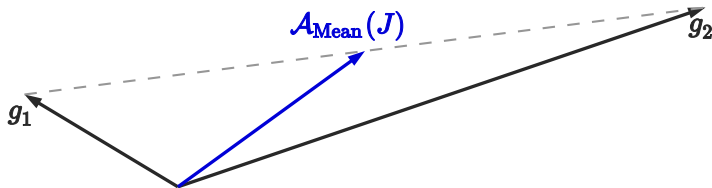
Unconflicting projection of gradients (UPGrad)

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$

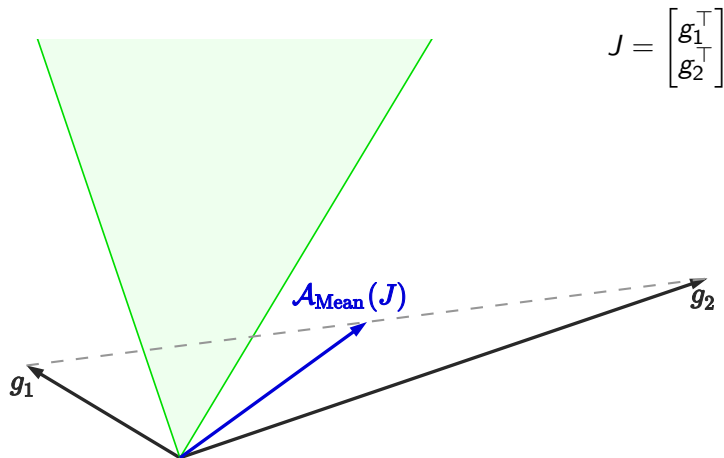


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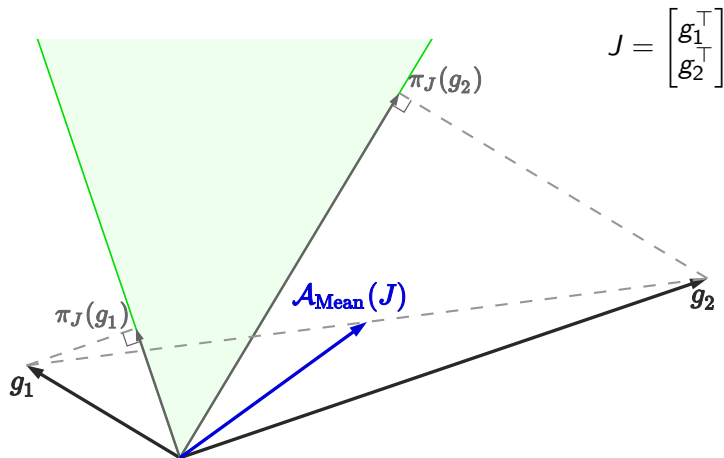
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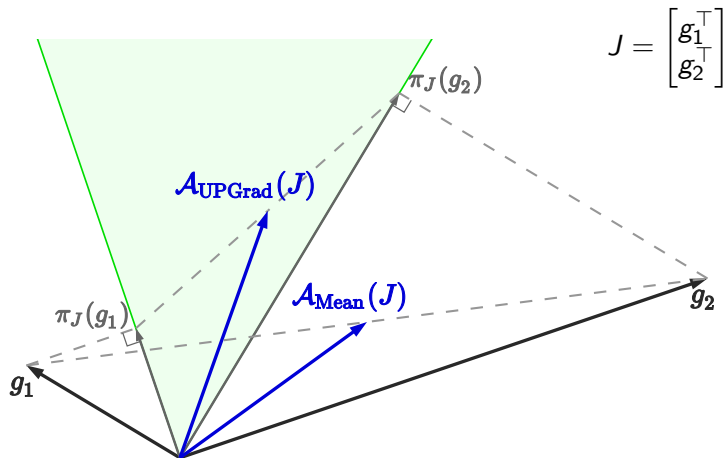
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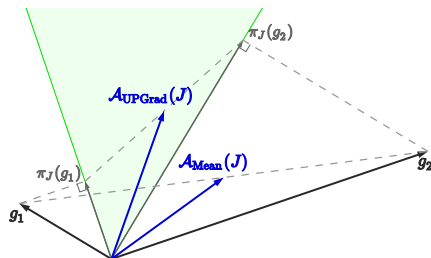


Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\mathbf{y} \in \mathbb{R}^n : \mathbf{0} \preceq J\mathbf{y}\}$.

$$\pi_J(\mathbf{x}) = \arg \min_{\mathbf{y} \in C_J} \|\mathbf{x} - \mathbf{y}\|^2$$

$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J(\mathbf{g}_i)$$



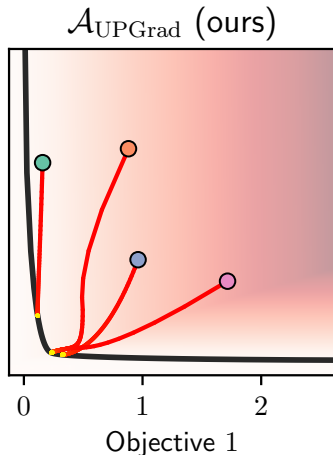
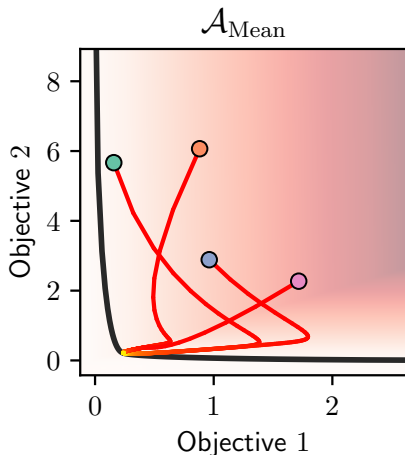
Applications and experimentation

Optimization trajectories

Goal: Minimize some convex quadratic function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, with conflicting gradients.

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Optimization trajectories: code

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

x = tensor([0., 0.], requires_grad=True)

for i in range(n_iters):
    y = f(x) # shape: [2]
    autojac.backward(y)
    # x now has a .jac field
    autojac.jac_to_grad([x], UPGrad())
    # x now has a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full project available at github.com/TorchJD/trajectories

Per-sample JD

Each element of the batch has its own loss.

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```
from torchjd import autojac
from torchjd.aggregation import UPGrad

batch_size = 16
model = Linear(10, 1)

for x, y in dataloader:
    z = model(x).squeeze(dim=1) # shape: [16]
    losses = loss_fn(z, y) # shape: [16]
    autojac.backward(losses)
    # parameters now have a .jac field
    autojac.jac_to_grad(model.parameters(), UPGrad())
    # parameters now have a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/iwrm.

Per-sample JD: results

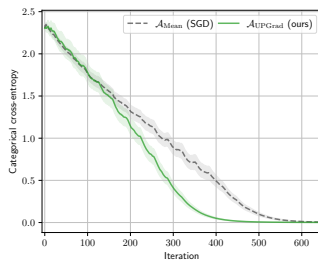
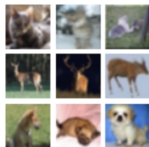
CIFAR-10



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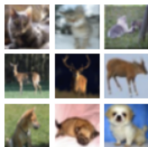
Training Loss

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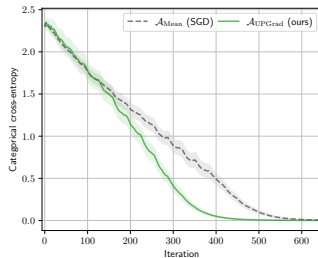


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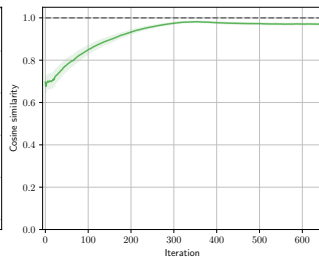
CIFAR-10



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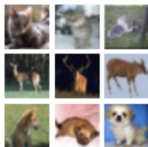


Update Similarity

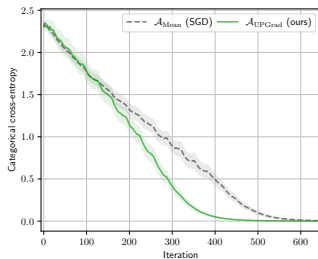


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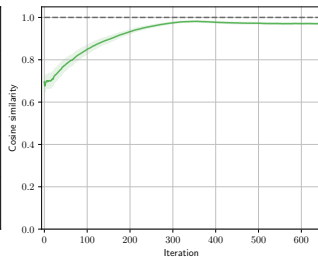
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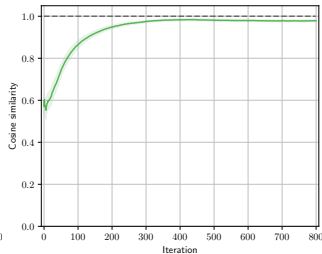
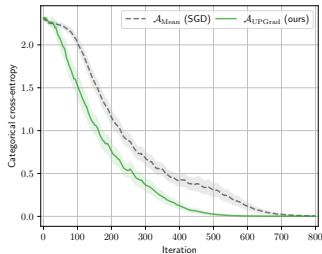
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SVHN

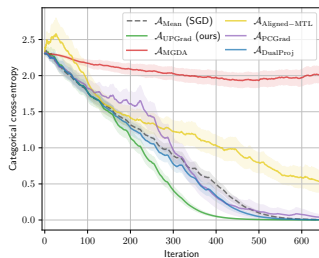


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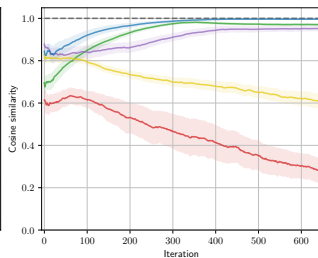
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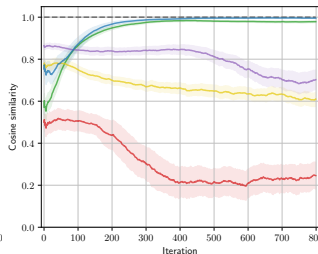
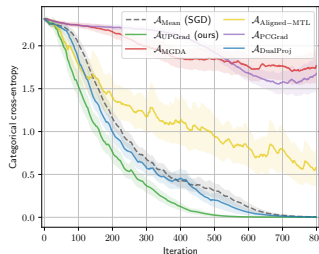
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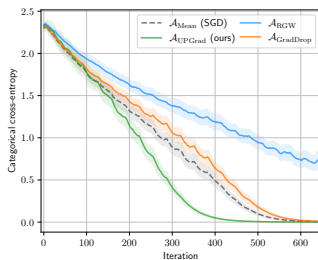


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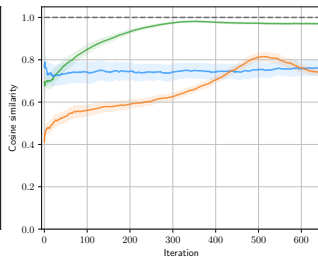
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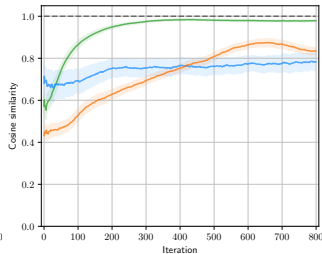
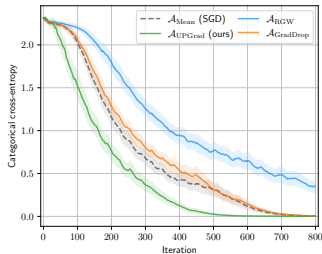
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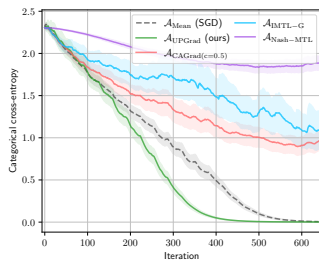


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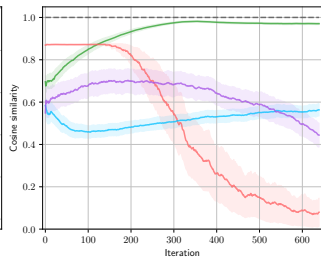
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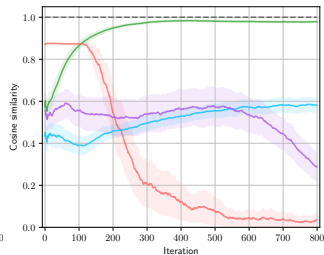
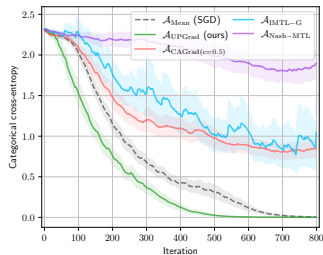
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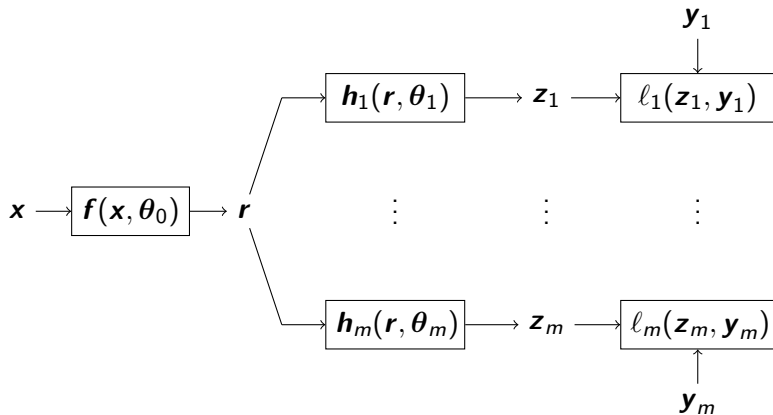
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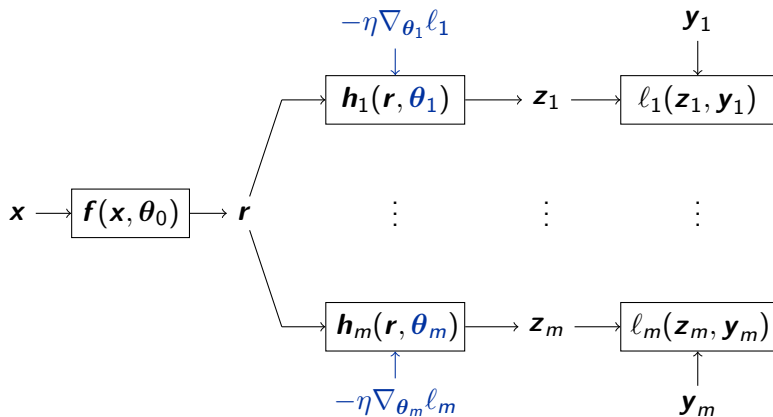
SVHN



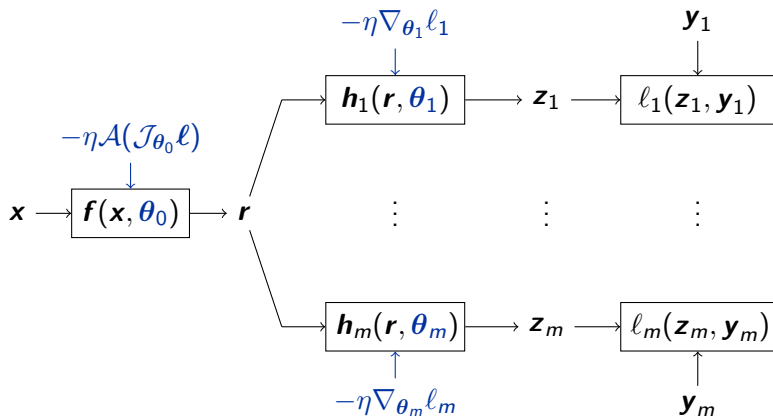
Multi-task learning



Multi-task learning



Multi-task learning



Multi-task learning: code

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

f = Linear(10, 5)
h1 = Linear(5, 1)
h2 = Linear(5, 1)
for x, y1, y2 in dataloader:
    r = f(x)
    z1, z2 = h1(r), h2(r)
    l1, l2 = loss_fn(z1, y1), loss_fn(z2, y2)
    autojac.mtl_backward([l1, l2], features=r)
    # head parameters now have a .grad field
    # backbone parameters now have a .jac field
    autojac.jac_to_grad(f.parameters(), UPGrad())
    # All parameters now have a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/mtl.

Conclusion

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Can JD solve problems that GD can't?

Appendix

Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\mathbf{f}(\mathbf{x}))$ is a linear combination of the rows of $\mathcal{J}\mathbf{f}(\mathbf{x})$.

Gramian-based JD

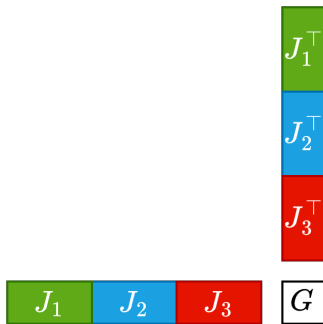
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This saves a lot of memory, so it's often much faster.

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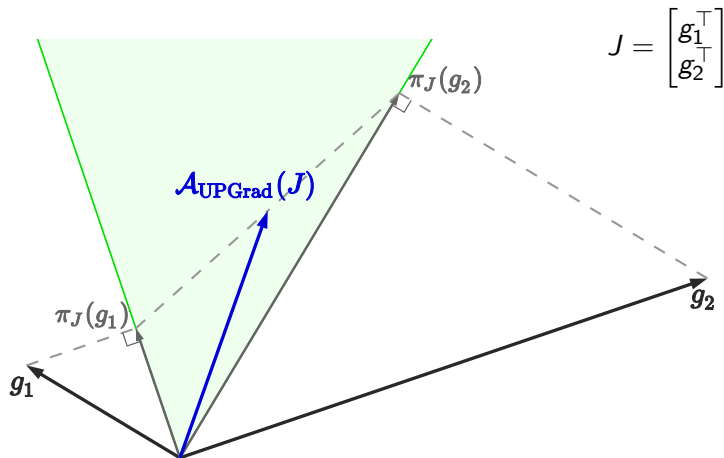
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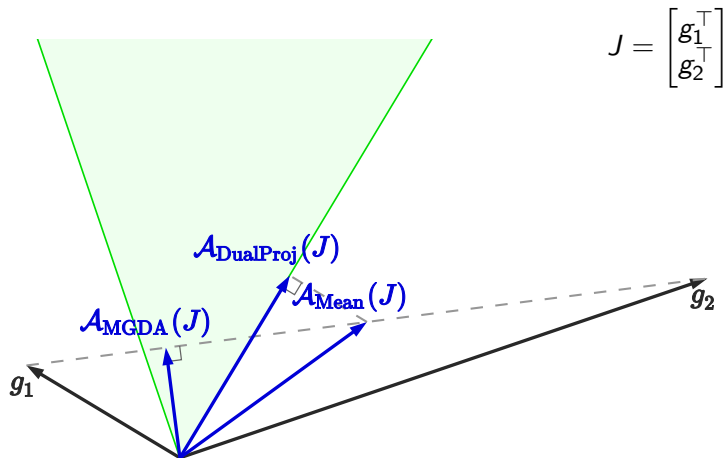
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Applicable to most other aggregators from the literature.

Other aggregators



Other aggregators

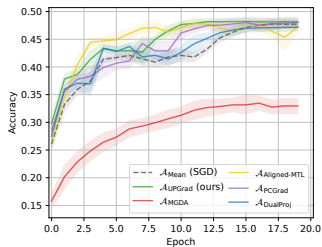


Per-sample JD: test accuracies

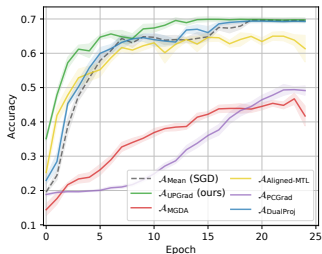
CIFAR-10



Test Accuracy



SVHN

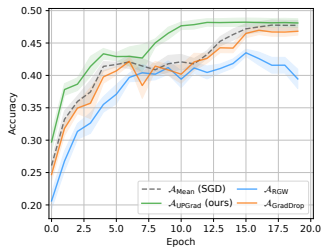


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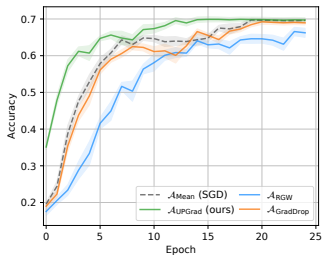
CIFAR-10



Test Accuracy

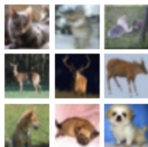


SVHN

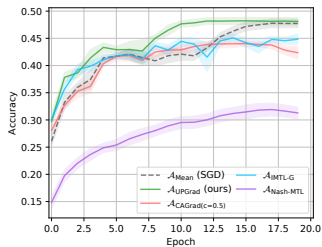


Per-sample JD: test accuracies

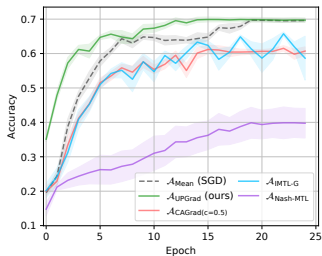
CIFAR-10



Test Accuracy

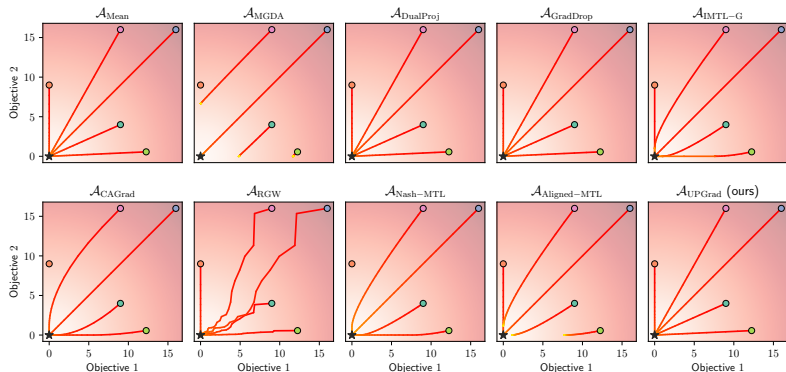


SVHN



Element-wise quadratic function trajectories

Minimize $\mathbf{f} : \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.



Convex quadratic form trajectories

Some convex quadratic function $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, with conflicting gradients.

